6.1 Experiment 1: complete particle–mesh operator action

Experiment 1 asks whether the operator that the integrator actually applies, $X \mapsto \Gamma_h X$, reproduces the periodic friction operator Γ of the screened longitudinal kernel $K(\mathbf{r}) = e^{-\kappa r} r^{-1} P(\mathbf{r})$, $\kappa = 1$, with the error predicted by the choice of (ξ, r_c, k_c, h, p). We follow the hierarchy used for the Landau particle method: static configurations first, so that no time-discretization error enters; then each source of error on its own; then the complete action. The model is not changed for the test. The friction is fully periodic, the $k = 0$ Fourier mode is retained in both the reference and the particle–mesh operator, and there is no free-space truncation and no energy null space. The only null space is the translation space $\mathcal{T} = \mathrm{span}\{T_\alpha\}_{\alpha=1}^{3}$, $T_\alpha = \mathbf{1}_N \otimes \mathbf{e}_\alpha$.

Why small dense-reference systems. Every error in this experiment is measured against a dense $3N \times 3N$ matrix Γ_{ref} and its full spectrum. The reference therefore has to be assembled pair by pair over many periodic images, and it has to be diagonalized. This restricts the experiment to $N \leq 256$. These sizes are chosen to make the reference computable; they are not a regime of interest in their own right. The two original test boxes are case A ($N = 30$, $L = 6$) and case B ($N = 40$, $L = 10$). Here L is the edge length of the periodic cube $[0, L)^3$, not a number of boxes. Their number densities are $n = N/L^3 = 0.139$ and 0.040, and $\kappa L = 6$ and 10. Both are single draws of independent uniform positions (seeds 7 and 13), with no minimum distance imposed; the closest pairs are at 0.395 and 0.589. Case B was taken large enough that the smallest nonzero lattice wave number $2\pi/L$ lies below the wave number at which the longitudinal eigenvalue of $\hat{K}^L$ changes sign. No physical argument selected $L = 6$ for case A: it is a small box in which the dense reference is cheap. The parameter sets used later in the paper were accepted when they met their tolerance on these two boxes. Two boxes are not evidence that the same parameters hold for other configurations, and we use them only as the starting point.

Configurations. Table 1 lists all configurations, including the shortest periodic (minimum-image) pair distance $d_{\min}$ of each. Since $\kappa = 1$, the column L also gives κL. The main set has $N = 128$ at the density of case A, $n_A = 30/216$, so $L = 9.732$. It contains three families, each with three independent seeds:

- *uniform*: independent uniform positions with no minimum distance;
- *perturbed lattice*: a bcc lattice of 4^3 cells ($N = 128$; an fcc lattice of 4^3 cells for $N = 256$), every site displaced by an isotropic Gaussian with $\sigma = 0.1\, d_{\mathrm{nn}}$, plus a random global shift. Here $d_{\mathrm{nn}} = 2.107$ (bcc) or 2.167 (fcc) is the nearest-neighbour distance;
- *clustered*: $N/16$ cluster centres drawn uniformly, and 16 particles per cluster drawn from an isotropic Gaussian of width $\sigma_c = 0.5$ around their centre.

For the clustered family, a draw is rejected if it comes closer than 0.25 to a particle already placed. This avoids near-coincident pairs, whose $1/r$ kernel would make $\|\Gamma_{\mathrm{ref}}\|_2$ arbitrarily large. The realised $d_{\min}$ is 0.251–0.259 ($N = 128$ and 256), so the constraint is active. For the uniform family, which has no constraint, $d_{\min}$ is 0.142–0.411. One configuration per family at $N = 256$ (with $L = 12.26$ at the same density) is a smaller representative check. All random numbers come from seeded generators; the seeds and generator parameters are stored with the positions.

Two box-length sets are kept apart, because they change different things.

- In the *fixed-N* set, $N = 64$ and $L \in \{4, 6, 8, 10, 12, 16\}$. Changing L therefore also changes the density, from $n = 1$ to 0.0156.

Table 1: Static configurations of Experiment 1. L is the edge length of the periodic cube, $n = N/L^3$ the number density, $d_{\min}$ the smallest periodic pair distance (range over seeds), $\kappa = 1$. A and B are the original two dense-reference boxes.

set	generation	N	$L = \kappa L$	$n = N/L^3$	$d_{\min}$	seeds
A	uniform, seed 7	30	6	0.139	0.395	1
B	uniform, seed 13	40	10	0.040	0.589	1
U128	uniform i.i.d.	128	9.73	0.139	0.142–0.411	3
L128	bcc 4^3, $\sigma = 0.1\, d_{\mathrm{nn}}$	128	9.73	0.139	1.208–1.366	3
C128	clusters of 16, $\sigma_c = 0.5$, $d \geq 0.25$	128	9.73	0.139	0.251–0.255	3
U256	uniform i.i.d.	256	12.3	0.139	0.148	1
L256	fcc 4^3, $\sigma = 0.1\, d_{\mathrm{nn}}$	256	12.3	0.139	1.011	1
C256	clusters of 16, $\sigma_c = 0.5$, $d \geq 0.25$	256	12.3	0.139	0.259	1
box-N	uniform i.i.d.	64	4–16 (6 values)	0.016–1.00	0.090–0.723	3
box-n	uniform i.i.d.	30–240	6–12 (4 values)	0.139	0.055–0.924	3

- In the *fixed-density* set, $n = n_A$ and $L \in \{6, 8, 10, 12\}$, so $N = 30, 71, 139, 240$.

Each set uses three uniform seeds per L.

Reference and its convergence. Γ_{ref} is built from the unsplit kernel by a direct periodic lattice sum,

$$(\Gamma_{\mathrm{ref}} X)_i = \sum_{j \neq i} \sum_{\mathbf{m} \in \mathbb{Z}^3,\ |\mathbf{r}_{ij} + \mathbf{m}L| \leq R} K(\mathbf{r}_{ij} + \mathbf{m}L)\,(X_i - X_j).$$

It uses neither a minimum-image approximation nor any splitting, and because nothing is removed from the sum the $k = 0$ mode is included automatically. The image radius was increased through $R = 15, 20, \ldots, 45$. Over all 44 configurations, the largest change relative to $R = 45$ is 2.5×10^{-12} at $R = 30$, 2.0×10^{-14} at $R = 35$, and 4.3×10^{-16} at $R = 40$, measured in $\|\cdot\|_2 / \|\Gamma_{\mathrm{ref}}\|_2$. The change decreases roughly like the tail $e^{-\kappa R}$ until it reaches round-off, and we take $R = 45$ as the reference. These differences are numerical evidence that the image sum has converged, not a rigorous bound on its truncation error: we have no analytic remainder estimate for the operator norm. The ladder also shares its inner shells, so it says nothing about round-off in the common part of the sum. That round-off is checked by the independent split construction below, which reproduces Γ_{ref} on the eleven main and legacy configurations for $\xi = 0.5$–1.4 to 1.4×10^{-16}–1.1×10^{-15}. Both levels are far below the smallest approximation error measured against the reference in this section, 2.3×10^{-10}. The reference is symmetric to 1.0×10^{-16} and annihilates the translations to 1.9×10^{-16}, both relative.

Split identity and the floor of the Fourier comparison. On nonzero separations, θ_S is evaluated in closed form and θ_L from its own s-integral (not as $e^{-\kappa r}/r - \theta_S$). The identity $\theta_S + \theta_L = e^{-\kappa r}/r$ (Eqs. (7)–(8)) then holds to 2.5×10^{-15} relative error. This was checked for $\xi \in \{0.5, 0.7, 1, 1.4, 2\}$, on a radial grid $10^{-3} \leq r \leq 20$, and on every pair image with $r \leq 12$ of the main and legacy configurations. The coefficients are accurate as well. With the Gauss–Legendre rules for $A(k)$, $B(k)$ and θ_L refined (twice the panels, 30 nodes per panel), these change by at most 1.6×10^{-15}, relative to $\|\hat{K}^L(k)\|_2$ or to $e^{-\kappa r}/r$, for $\xi \in \{0.5, 0.7, 1, 1.4, 2\}$, $k \leq 14\xi$ and $10^{-3} \leq r \leq 20$. The quadrature of θ_S agrees with its closed form to 1.0×10^{-15}.

At operator level we combine two independent constructions:

- Γ_S from the real-space lattice sum of θ_S over $|\mathbf{r}+\mathbf{m}L| \le 45$;
- Γ_L^{F} from the mesh-free lattice Fourier sum $V^{-1}\sum_{|\mathbf{k}|\le k_c} \hat{K}^L(\mathbf{k})\cos(\mathbf{k}\cdot\mathbf{r})$, including $\mathbf{k}=0$, with the finite cutoff $k_c = 2\xi \cdot 6.5$. In the Fourier-cutoff sweep below, the truncation error falls below the fit floor defined below by $k_c/(2\xi) = 5.5$ and reaches the numerical floor of the comparison near $k_c/(2\xi) = 6$, well before this cutoff.

On the eleven main and legacy configurations and for $\xi \in \{0.5, 0.7, 1, 1.4\}$, their sum reproduces Γ_{ref} to between 1.4×10^{-16} and 1.1×10^{-15} (Fig. 1a).

An earlier version of this computation stopped at 6.7×10^{-15}–2.2×10^{-13}, and the Fourier-cutoff sweep levelled off at 1.2×10^{-13} ($\xi=0.7$) and 1.6×10^{-13} ($\xi=1$). The cause was an artificial rounding in the coefficient routine: $A(k)$ and $B(k)$ were evaluated at $|\mathbf{k}|$ rounded to twelve decimal places, which perturbs the kernel input by up to 5×10^{-13}. The routine now evaluates them at the unrounded $|\mathbf{k}|$ (the float64 norm of $\mathbf{k}$). All Fourier-side quantities of this section, including the production operator, were recomputed with the corrected routine. A recorded check (Table 2; case A, a perturbed lattice and a uniform configuration at $\xi = 0.7$ and 1) compares the Fourier sum Γ_L^{F} with the real-space image sum Γ_L and separates four sources:

- *wave-number rounding*: with $|\mathbf{k}|$ rounded to twelve decimals the floor is 2.3×10^{-14}–1.5×10^{-13}; with the unrounded $|\mathbf{k}|$ it is 6.3×10^{-17}–7.6×10^{-16}, lower by a factor 130–370 in every checked case. Evaluating the coefficients on integer shells $|\mathbf{m}|^2$, with k computed from the shell, gives 5.4×10^{-17}–8.5×10^{-16}, the same level;
- *numerical quadrature*: refining the Gauss–Legendre rules for $A(k)$ and $B(k)$ gives 6.5×10^{-17}–7.9×10^{-16}, again the same level. The θ_L quadrature inside the real-space image sum Γ_L ($\xi r < 1.5$) was not refined in this check; θ_L itself changes by at most 1.6×10^{-15} under refinement (above);
- *floating-point arithmetic*: evaluating the Fourier sum (phases, cosines and accumulation) in 80-bit arithmetic from the same float64 inputs lowers the floor only by a factor 1.0–1.9, to 3.5×10^{-17}–4.0×10^{-16}; the float64 and 80-bit sums differ by 4.2×10^{-17}–6.7×10^{-16}. Float64 arithmetic in the Fourier sum is therefore part of the remaining floor, but not all of it;
- *truncation of the real-space reference*: the real-space image sum Γ_L changes by 4.1×10^{-17}–4.3×10^{-16} between $R=40$ and $R=45$. This change contains the shell $40 < r \le 45$ and the round-off of the image sum, so it estimates the numerical uncertainty of the reference rather than its truncation alone.

The earlier precision check, which compared float64 with 80-bit evaluation and with refined coefficients, could not see the rounding, because all of its variants used the same rounded coefficients. With the unrounded $|\mathbf{k}|$, the remaining floor is at most 7.6×10^{-16}. It has the size of the float64 round-off of the Fourier sum and of the numerical uncertainty of the real-space image sum. The image sum itself (float64 evaluation, θ_L quadrature) was not recomputed in higher precision, so we do not separate these contributions further. It lies two to three orders of magnitude below the earlier artificial floor, and far below every error used in a fit ($\ge 1.1\times10^{-13}$) and every production error ($\ge 2.3\times10^{-10}$).

Real-space and Fourier cutoffs. Each truncation is measured with no other approximation present:

- $\|\Gamma_S^{r_c} - \Gamma_S\|_2$, where Γ_S is the real-space lattice sum of θ_S over $|\mathbf{r}+\mathbf{m}L| \le 45$;

Table 2: Floor of the comparison between the mesh-free Fourier sum Γ_L^{F} ($k_c = 2\xi \cdot 6.5$) and the real-space image sum Γ_L ($|\mathbf{r} + \mathbf{m}L| \le 45$), $\|\Gamma_L^{\mathrm{F}} - \Gamma_L\|_2 / \|\Gamma_{\mathrm{ref}}\|_2$, range over case A, a perturbed lattice and a uniform configuration ($N = 128$, seed 1) at $\xi = 0.7$ and 1.

variant	range	source probed
$\lvert\mathbf{k}\rvert$ rounded to 12 decimals (earlier)	2.3×10^{-14}–1.5×10^{-13}	wave-number rounding
unrounded $\lvert\mathbf{k}\rvert$, float64 (current)	6.3×10^{-17}–7.6×10^{-16}	baseline
k from the integer shell $\lvert\mathbf{m}\rvert^2$, float64	5.4×10^{-17}–8.5×10^{-16}	wave-number input
unrounded $\lvert\mathbf{k}\rvert$, refined quadrature	6.5×10^{-17}–7.9×10^{-16}	quadrature
unrounded $\lvert\mathbf{k}\rvert$, sum in 80 bits	3.5×10^{-17}–4.0×10^{-16}	floating point
float64 sum − 80-bit sum	4.2×10^{-17}–6.7×10^{-16}	floating point
image sum Γ_L: $R = 40$ against $R = 45$	4.1×10^{-17}–4.3×10^{-16}	reference: tail + round-off

- $\|\Gamma_L^{k_c} - \Gamma_L\|_2$, where $\Gamma_L^{k_c}$ is the mesh-free lattice Fourier sum over $|\mathbf{k}| \le k_c$, and Γ_L is the real-space lattice sum of θ_L over $|\mathbf{r} + \mathbf{m}L| \le 45$ ($\theta_L = e^{-\kappa r}/r - \theta_S$ for $\xi r \ge 1.5$, its s-integral otherwise).

Both are normalized by $\|\Gamma_{\mathrm{ref}}\|_2$ (Fig. 1b,c).

All fits in this section use one exclusion rule. Let $\varepsilon_{\mathrm{num}}$ be the largest numerical discrepancy found above: the operator-level split identity and every row of Table 2 except the rounded-$|\mathbf{k}|$ one (the Fourier-sum variants, the float64 against 80-bit difference and the change of the image sum between $R = 40$ and 45). This gives $\varepsilon_{\mathrm{num}} = 1.1 \times 10^{-15}$. A point enters a fit only if its error is at least $100\,\varepsilon_{\mathrm{num}} = 1.1 \times 10^{-13}$, so that the floor contributes at most 1% to it. Points below this level stay in the data files. In Fig. 1b,c they lie below the dotted fit-floor line. The two excluded meshes of the mesh scans are listed below and in the summary file. The earlier threshold of 10^{-12} was chosen to stay above the artificial floor; it is replaced by this rule.

For the cutoffs we fit $\log E = a - cx^2$ with $x = \xi r_c$ or $x = k_c/(2\xi)$, using only $x \ge 1$ and points above the fit floor. Steps at which a larger cutoff admits no new pair image or lattice shell are also left out (see below). At $\xi = 0.7$ the fitted Gaussian rates are:

- real space: median $c = 0.99$, range 0.98–1.02 over the eleven main and legacy configurations;
- Fourier space: median $c = 0.97$, range 0.96–1.00.

At $\xi = 1$ the medians are again 0.99 (range 0.98–1.03) and 0.98 (0.96–0.99). The $N = 256$ configurations give 0.98–1.01 (real space) and 0.97–1.00 (Fourier). The fit windows end at $x = 5.25$–5.5 (real space) and 5–5.25 (Fourier). Removing the continuum prefactor of the tails (Eq. (17), $E \propto x\,e^{-x^2}$) by dividing by x overcorrects, giving medians of 1.04–1.05 (real space) and 1.03 (Fourier) at the two values of ξ. Neither form gives $c = 1$ within the fit window: over this window the operator-norm error is not described by the continuum prefactor x alone, and its effective prefactor lies between x^0 and x^1. We do not interpret c as a measured universal exponent or as an asymptotic prefactor.

Below the fit floor the Fourier error falls until it levels off near $x = 6$, in every configuration, at 2×10^{-17}–3.6×10^{-16} ($\xi = 0.7$) and 4×10^{-17}–6.7×10^{-16} ($\xi = 1$), the level of Table 2. The real-space error keeps falling in 25 of the 28 curves, to at most 9.2×10^{-17} ($\xi = 0.7$) and 4.0×10^{-17} ($\xi = 1$) at $x = 6.5$; only three curves flatten. Its comparison involves two real-space sums over the same images, so it has no comparable floor. The plateau near 10^{-13} seen with the earlier coefficient routine is gone. The finite particle set makes the curves staircases at small cutoffs:

- on the perturbed lattices, 13 real-space steps with $\xi r_c \le 1.25$ admit no new pair image;

- for case A, 3 Fourier steps with $k_c/(2\xi) \leq 1$ admit no new lattice shell.

These points keep their equal errors and are flagged in the data, not smoothed. The seven of them that fall inside a fit window are excluded from the fits.

Mesh spacing, B-spline order and aliasing. With $\xi = 0.7$ and the production Fourier cutoff $k_c = 5.74$ ($s = 4.1$), we compare the particle–mesh long-range operator with the mesh-free lattice sum over the same modes. The difference $\Gamma_L^{\mathrm{mesh}} - \Gamma_L^{k_c}$ is then the mesh error alone: both sides retain the same modes, so the Fourier truncation cancels. For the eleven main and legacy configurations the scan covers $p = 3, \ldots, 8$ and nine FFT-friendly meshes from $\eta = k_c h/\pi \approx 0.91$–$0.99$ down to 0.23–0.25. Each mesh is evaluated at eight grid shifts: the unshifted grid and seven random global offsets in $[0, h)^3$ (Fig. 1d–f). The $N = 256$ checks use $p = 5, 7$, six meshes (η down to 0.23) and four shifts. The scan was repeated at the larger cutoff $s = 5.5$ ($e^{-s^2} \approx 7 \times 10^{-14}$), which retains more modes near the mesh limit, for $p = 4, 6, 8$, seven meshes and eight shifts, on cases A and B and seed 1 of each $N = 128$ family. Adjacent meshes differ by 7–41% in h.

We fit the RMS over shifts on the four finest meshes whose error is above the fit floor. This rule is the same for every configuration, p and cutoff; it targets the smallest h available without letting the floor into the fit. At $s = 4.1$ no mesh falls below the fit floor; the smallest error is 4.5×10^{-13}. At $s = 5.5$ the finest mesh for $p = 8$ does so in two configurations (case A and uniform seed 1, $\eta \approx 0.25$, error down to 8.9×10^{-14}); there the rule takes the next four meshes. The fit windows lie between $\eta = 0.23$ and 0.50 at $s = 4.1$ (typically 0.25–0.40) and reach up to $\eta = 0.61$ at $s = 5.5$. The fitted slopes in h are, as medians (with ranges for the four finest meshes) over the eleven main and legacy configurations ($s = 4.1$) and over the five configurations of the $s = 5.5$ scan:

	$s = 4.1$ (production k_c, 11 configurations)			$s = 5.5$ (5 configurations)
p	four finest, median (range)	all above floor	$\eta \geq 0.5$	four finest, median (range)
3	3.09 (2.90–3.22)	3.14	3.31	–
4	4.07 (3.85–4.36)	4.26	4.53	4.10 (3.97–4.29)
5	5.24 (4.88–5.43)	5.50	5.97	–
6	6.19 (6.13–6.44)	6.72	7.30	6.25 (6.16–6.35)
7	7.46 (7.03–7.66)	7.97	8.68	–
8	8.50 (8.42–8.68)	9.24	10.18	8.51 (8.38–8.72)

For $p = 5, 7$ the three $N = 256$ checks give 5.12–5.39 and 7.44–7.58, inside these ranges. On the four finest meshes the slopes are thus consistent with the h^p trend predicted by the alias analysis (Eqs. (54), (58)). A fit over a finite set of meshes cannot establish the asymptotic order itself, and we do not claim that it does. The median slopes lie above p for every p, by 0.07–0.09 for $p = 3, 4$ and by up to 0.5 for $p = 8$. Individual fits fall below p in seven of the 66 fits at $s = 4.1$ (down to 2.90, 3.85 and 4.88 for $p = 3, 4, 5$) and in two of the 15 at $s = 5.5$ (3.97 and 4.00, $p = 4$). They grow when coarser meshes are included: over every mesh above the floor, and even more on the coarse meshes alone ($\eta = 0.55$–0.99, where they are 1.10–$1.27\,p$). This is consistent with the leading alias image. Its factor q^p, with $q = \eta_k/(2 - \eta_k)$ and $\eta_k = kh/\pi \leq \eta$ for a retained mode k, has local exponent between p and $2p/(2 - \eta)$ in h, and approaches p only as $\eta \to 0$. Slopes between adjacent meshes are too noisy to show this directly. We therefore describe the mesh error as consistent with the predicted h^p trend for $\eta \approx 0.23$–0.50, with faster pre-asymptotic decay on coarser meshes, and we report no asymptotic constant.

The phase sensitivity is moderate. Over all eight-shift scans, the largest error over shifts is at most 1.30 times the RMS and at most 1.92 times the smallest. Fitted on the same windows, the

Table 3: Parameter sets of the production action test (accepted at $\xi = 0.7$ for the stated tolerance on cases A and B). M is the next $2, 3, 5$-smooth mesh size with $h \le \eta_N \pi / k_c$; the realised $\eta = k_c h/\pi$ is therefore at most η_N and depends on L.

set	ξ	s	$r_c = s/\xi$	$k_c = 2\xi s$	η_N, p	M (A / B / N=128 / N=256)
T5	0.7	3.494	4.991	4.891	0.8, 6	12 / 20 / 20 / 24
T7	0.7	4.100	5.857	5.740	0.7, 7	16 / 27 / 27 / 36
T9	0.7	4.628	6.611	6.479	0.6, 8	24 / 36 / 36 / 45

slope of the maximum differs from that of the RMS by -0.26 to $+0.20$. On the unshifted grid, the complete off-grid decomposition

$$K_h - K_F = V^{-1} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}_{ij}} \left[D_h + H_h(\alpha_i^* + \alpha_j + \alpha_i^* \alpha_j) \right]$$

reproduces the particle–mesh pair response to 3×10^{-15} of $\max |K_F|$. The literal spread–FFT–influence–gather response, computed one source particle at a time, agrees with the mode-space pair response to 1.3×10^{-15}. We checked this for $p = 4, 7$ on two meshes ($\eta \approx 0.46$–0.49 and 0.91–0.99) for case A, case B and seed 1 of each $N = 128$ family; these meshes lie above the fit windows of the slope fits. In the decomposition:

- the linear cross terms $\alpha_i^* + \alpha_j$ between the principal term and the aliases carry 0.996–1.003 of the mesh error (spectral norm);
- the alias–alias term carries at most 0.9%;
- D_h is at most 4.3×10^{-9} of the error, as it should with ideal deconvolution.

An estimate that kept only the alias–alias term would therefore miss the mesh error almost completely.

The complete production action. We now apply the production code path itself: the real-space pair loop, B-spline spreading of the three components, FFT, multiplication by the influence function $\hat{K}^L/|\hat{W}|^2$ on $|\mathbf{k}| \le k_c$, inverse FFT, adjoint gather, and the degree correction. We use it in two ways:

- applied to the $3N$ unit vectors, it gives Γ_h column by column, without any hand-assembled matrix;
- applied to probe vectors, it gives the action directly.

Three parameter sets were tested (Table 3): the cheapest sets accepted at $\xi = 0.7$ for the tolerances 10^{-5}, 10^{-7} and 10^{-9} on cases A and B (T5, T7, T9). The literal columns agree with an independent mode-space assembly to 1.1×10^{-15}. That assembly, in turn, equals an independently built real-space cutoff sum $\Gamma_S^{r_c}$ plus the same mesh pair matrix Γ_L^{mesh} to 1.2×10^{-15}. The corresponding error matrices therefore add up to $\Gamma_h - \Gamma_{\mathrm{ref}}$, up to residuals of order 10^{-15} (the split identity and the two agreements above). With the budget terms $\varepsilon_S = \|\Gamma_S^{r_c} - \Gamma_S\|_2$, $\varepsilon_F = \|\Gamma_L^{k_c} - \Gamma_L\|_2$ and $\varepsilon_{\mathrm{mesh}} = \|\Gamma_L^{\mathrm{mesh}} - \Gamma_L^{k_c}\|_2$, all normalized by $\|\Gamma_{\mathrm{ref}}\|_2$, this gives $\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2 / \|\Gamma_{\mathrm{ref}}\|_2 \le \varepsilon_S + \varepsilon_F + \varepsilon_{\mathrm{mesh}}$ up to those residuals; here the sum is 1.2–2.0 times the production error. All three sets keep $\mathbf{k} = 0$ and exclude no influence mode.

We report three different error measures and keep them apart:

- the *relative operator error* $\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2/\|\Gamma_{\mathrm{ref}}\|_2$, the quantity on which T5, T7 and T9 were accepted;
- the *relative action error* $\|(\Gamma_h - \Gamma_{\mathrm{ref}})X\|/\|\Gamma_{\mathrm{ref}}X\|$ for a given vector X;
- for the translations, where $\Gamma_{\mathrm{ref}}X$ vanishes up to round-off, the *operator-normalized residual* $\|\Gamma_h X\|/(\|\Gamma_h\|_2\|X\|)$ and the error $\|(\Gamma_h - \Gamma_{\mathrm{ref}})X\|/(\|\Gamma_{\mathrm{ref}}\|_2\|X\|)$, instead of a ratio with a vanishing denominator.

Figure 2a shows the relative operator error. Cases A and B meet their tolerances, as they were selected to. The other configurations do not all meet them. T5 and T9 were tested on the 14 configurations of the legacy, main and check sets, and T7 also on the 30 box-length configurations. The relative operator error exceeds the tolerance in 13 of these 72 cases:

- T5: the three perturbed lattices at $N = 128$ (1.16–1.29×10^{-5}) and the lattice at $N = 256$ (1.17×10^{-5});
- T7: the same four lattices (1.94–2.11×10^{-7} at $N = 128$, 1.09×10^{-7} at $N = 256$) and one uniform box of the fixed-density set, $N = 30$, $L = 6$, seed 3 (1.40×10^{-7});
- T9: the four lattices (1.96–2.55×10^{-9}).

The excess is up to a factor 2.5. The uniform and clustered configurations of the main and check sets stay within the tolerances. The exceedances are not specific to lattices, however: the uniform box that exceeds T7 has the same N, L and generator as case A and differs from it only in the seed.

Figure 2b shows where the excess comes from. Both the numerator and the denominator of the relative error depend on the configuration. At T7 the absolute error $\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2$ lies in 5.7×10^{-8}–5.5×10^{-7} over all 44 configurations. It is 3.4–5.2×10^{-7} for the clustered and 1.0–1.5×10^{-7} for the perturbed-lattice configurations, and in the fixed-N set it is larger at higher density, though not monotonically (see below). Meanwhile $\|\Gamma_{\mathrm{ref}}\|_2$ ranges from 0.72 to 35. Close pairs strongly influence it, but the nearest pair alone does not set it. With $\theta(r) = e^{-\kappa r}/r$, the ratio $\|\Gamma_{\mathrm{ref}}\|_2/\theta(d_{\min})$ is 2.0–4.1; an isolated pair gives exactly 2. The ratio reaches 2.6–4.1 on the lattices and 3.2–3.7 on the clustered configurations, where many pairs at similar distances contribute. Every configuration that exceeds its tolerance has a small norm: 0.72–0.94 for the lattices, and 1.50 for the uniform box, whose closest pair is at 0.92. Their absolute errors are moderate (1.0–2.1×10^{-7} at T7). The uniform box and case A have nearly the same absolute error at T7 (2.1×10^{-7}), but case A's norm is 3.94, so its relative error is 5.4×10^{-8} and the box's is 1.40×10^{-7}. A small norm is not sufficient for an exceedance either: the fixed-N box with $L = 16$, seed 1, has $\|\Gamma_{\mathrm{ref}}\|_2 = 1.34$ and a relative error of 7.5×10^{-8}, because its absolute error is lower. A tolerance stated relative to $\|\Gamma_{\mathrm{ref}}\|_2$ and accepted on two configurations is therefore not guaranteed to hold on other configurations; here it is exceeded in 13 of the 72 configuration–set pairs.

For the main and legacy configurations at T7 the mesh term is the largest in the budget ($\leq 1.5 \times 10^{-7}$ relative), followed by the real-space and Fourier truncations ($\leq 1.05 \times 10^{-7}$ and $\leq 7 \times 10^{-8}$, largest for the lattices). The real-space term is the larger one for the $N = 256$ lattice and for five boxes at the smallest realised $\eta = 0.61$: the three fixed-N boxes with $L = 4$ and two of the fixed-density boxes with $L = 8$. At T5 it is the largest for the uniform and lattice configurations at $N = 128$.

The probes are applied through the production matvec (Fig. 2c):

- sixteen mean-free Gaussian vectors;

- the lowest nonzero periodic wave number $2\pi/L$ along each axis, longitudinal and transverse, cosine and sine, projected onto $\mathcal{T}^{\perp}$;
- the three translations.

At T7 the largest relative action errors on the main and legacy sets are 1.7×10^{-7} (Gaussian), 2.7×10^{-7} (longitudinal) and 4.2×10^{-7} (transverse), against a largest relative operator error of 2.1×10^{-7}. A relative action error can exceed the configuration's relative operator error by at most the factor $\|\Gamma_{\mathrm{ref}}\|_2\|X\|/\|\Gamma_{\mathrm{ref}}X\|$. For the long-wavelength probes this factor is large: they vary slowly over the kernel range $1/\kappa$, so the pair differences $X_i - X_j$ on which Γ acts are small, and $\|\Gamma_{\mathrm{ref}}X\|/(\|\Gamma_{\mathrm{ref}}\|_2\|X\|)$ is as small as 2.5×10^{-2} (longitudinal) and 1.1×10^{-2} (transverse) on the main and legacy sets. Every longitudinal and transverse probe there exceeds its configuration's relative operator error, by up to a factor 14 and 18; about half of the Gaussian probes do, by up to a factor 6.

Meeting the operator tolerance therefore does not mean meeting it for every action. At T7, 39 of the 44 configurations have a relative operator error at most 10^{-7}. Every one of these 39 has a transverse probe with relative action error above 10^{-7}, 32 have such a longitudinal probe, and 21 have such a Gaussian probe; cases A and B are among them for all three kinds. Over all 44 configurations the largest relative action errors at T7 are 2.2, 3.7 and 4.5 times the tolerance (Gaussian, longitudinal, transverse). T5 and T9 behave the same way on the 14 configurations tested with them. Of the 10 configurations within the operator tolerance, 8 (T5) and all 10 (T9) have a transverse probe above it, and among these the largest transverse action errors are 2.5 and 4.9 times the tolerance. For slow modes, the relative action error has to be checked directly, or bounded through the factor above. For the translations:

- the production $\Gamma_h T_\alpha$ is exactly zero in floating point, because the degree correction cancels the constant field term by term;
- $\|\Gamma_{\mathrm{ref}}T_\alpha\|/(\|\Gamma_{\mathrm{ref}}\|_2\|T_\alpha\|) \le 1.9 \times 10^{-16}$.

The literal Γ_h is symmetric to 1.0×10^{-16} (relative Frobenius norm). For Experiment 3 we record the smallest eigenvalue on $\mathcal{T}^{\perp}$. It ranges over 8×10^{-4}–1.92 across the configurations, and Γ_h changes it by at most 4.2×10^{-5} (T5), 1.6×10^{-6} (T7) and 2.6×10^{-8} (T9) relative. In every case $\|\Gamma_h - \Gamma_{\mathrm{ref}}\|_2 < \lambda_{\min}(\Gamma_{\mathrm{ref}}|_{\mathcal{T}^{\perp}})$, with margin at least $0.997\,\lambda_{\min}$.

Box length, density and system size. At T7, the two box-length sets give different information (Fig. 2d). All values below are medians over the three seeds.

Fixed $N = 64$. Here $n = N/L^3$ and κL change together, so this set does not separate density from box length. Because the kernel range $1/\kappa$ is fixed, and the fixed-density set below changes the error by at most a factor 2.2, we read the larger change here mainly as a density effect. The absolute error falls from 5.3×10^{-7} ($L = 4$, $n = 1$) to 1.0×10^{-7} ($L = 16$, $n = 0.016$). Between $L = 4$ and $L = 16$ all three budget terms decrease, though not monotonically:

- the absolute real-space term goes from 3.2×10^{-7} to 2.6×10^{-8} (sublinear in n);
- the Fourier term goes from 1.2×10^{-7} to 1.8×10^{-8}, with a rise from 2.3×10^{-8} ($L = 6$) to 4.1×10^{-8} ($L = 8$);
- the mesh term goes from 2.9×10^{-7} to 8.8×10^{-8}. It is the largest term for $L \ge 6$, and the realised η also modulates it.

Rounding M to a $2, 3, 5$-smooth size gives $\eta = 0.61$ at $L = 4$ and 8 but 0.65–0.69 otherwise. The dip at $L = 8$ coincides with the smaller η: for $p = 7$ the h^p trend predicts a factor $(0.61/0.68)^7 \approx 0.45$ in the mesh term, about the observed drop, but we did not rerun $L = 8$ at matched η.

Fixed density n_A. Here N grows with L ($N = 30$, 71, 139, 240), so this is a system-size check at fixed density; it does not isolate the effect of L. The absolute error is 2.1×10^{-7} ($L = 6$), 1.1×10^{-7} ($L = 8$, $\eta = 0.61$), 2.5×10^{-7} ($L = 10$) and 2.5×10^{-7} ($L = 12$). The dip at $L = 8$ again coincides with $\eta = 0.61$. At nearly equal η (0.68–0.69) the two larger systems lie about 20% above $L = 6$, beyond the seed spread (2.0–2.1 $\times 10^{-7}$ against 2.3–2.7 $\times 10^{-7}$). The difference is mostly in the mesh term (1.9 against 2.3×10^{-7}). Because N, L and the particle arrangement change together, this set cannot attribute the difference to L, and we did not isolate its cause. Over $L = 6$–12 the absolute error changes by at most a factor 2.2, most of it the drop at $L = 8$, which coincides with the smaller realised η. The relative error also inherits the configuration-dependent $\|\Gamma_{\mathrm{ref}}\|_2$, as above.

What the experiment supports. The original cases A and B show that the accepted parameter sets meet their relative tolerances on two uniform boxes. On their own they do not support a claim about other configurations. The supplementary experiment adds the following:

1. an independent reference built from the unsplit kernel, with $k = 0$, whose image-radius ladder shows numerical convergence to below 10^{-15} and which an independent split construction reproduces to 1.1×10^{-15} on the main and legacy configurations;

2. the split identity and the coefficient quadrature at round-off over a range of ξ, and an attribution of the former 10^{-13} floor to an artificial wave-number rounding, now removed;

3. Gaussian decay of both truncations, and mesh-error slopes consistent with the predicted h^p trend for $\eta \approx 0.23$–0.50, with similar rates across three configuration families, two particle numbers and grid shifts, and with the linear alias cross terms carrying 99.6–100.3% of the mesh error on the meshes checked ($\eta \approx 0.46$–0.99);

4. a verification of the production spread–FFT–influence–gather action itself, including null-space and symmetry properties.

It also shows a limitation of the original acceptance. T5, T7 and T9 meet their relative operator tolerance on cases A and B, but not on every other configuration. Of the 72 configuration–set pairs, 13 exceed it, by up to a factor 2.5: the perturbed lattices, and one uniform box with the size and density of case A. Even where the operator tolerance is met, the relative action error of slow modes exceeds it by up to a factor 4.5 (T7) and 4.9 (T9). All results are for $\kappa = 1$, $N \leq 256$ and static configurations.

Consequence for parameter selection and for Experiment 2. Parameter sets accepted on a few reference configurations carry no automatic accuracy guarantee for other configurations. Calibrating on the configuration with the smallest $\|\Gamma_{\mathrm{ref}}\|_2$ would not be sufficient either, because the absolute error in the numerator also changes with the configuration, by a factor of about ten at T7 here. Parameters therefore have to be verified over the range of configurations on which they are used. That can be done by dense-reference or refined-parameter checks on configurations sampled from the target ensemble, or by choosing the parameters from a reliable error bound. Where slow modes matter, the check has to include the action on them, not only the operator norm. The same qualification applies to the equal-accuracy timings of Experiment 2. They compare parameter sets

that meet the same relative tolerance on cases A and B, which is equal accuracy on those two configurations, not a guaranteed accuracy on the configurations a simulation visits. Experiment 2 was not rerun for this section.

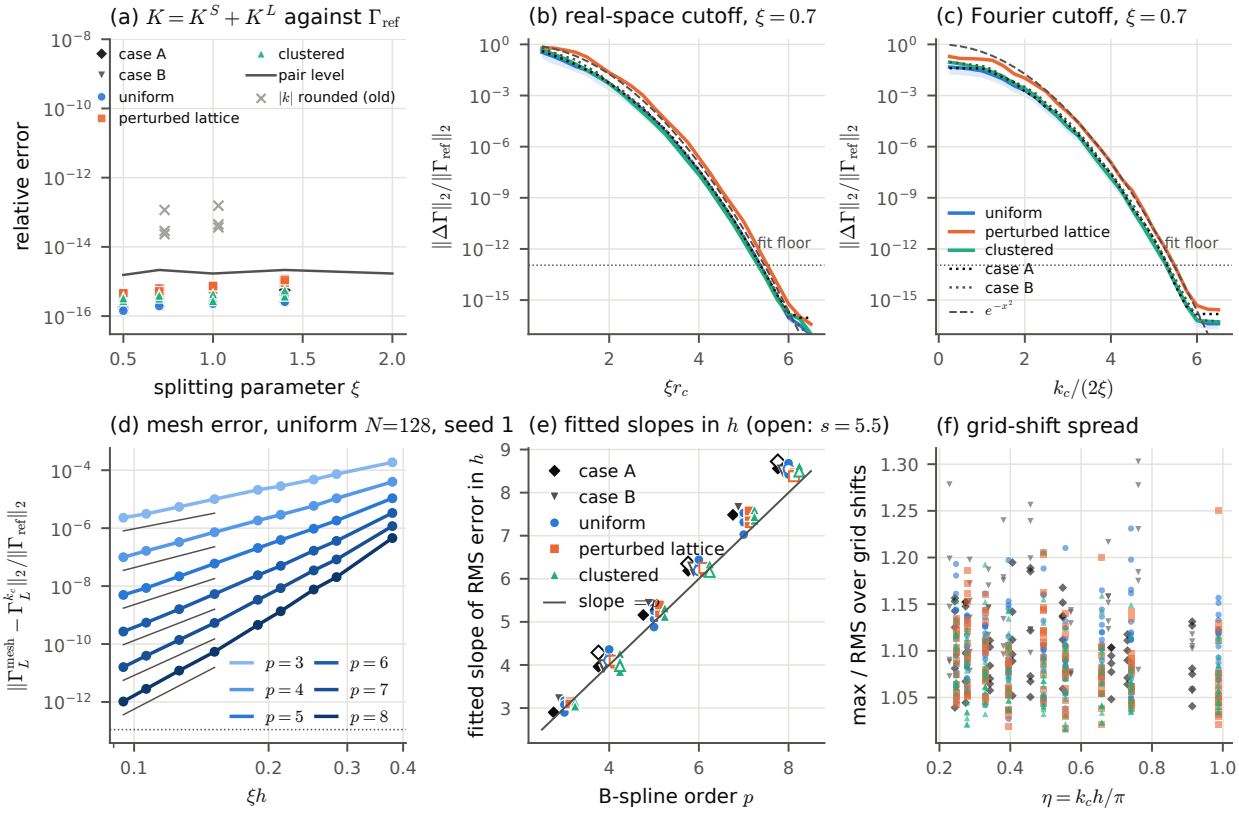


Figure 1: Error sources of Experiment 1, $\kappa = 1$. (a) Split identity: Γ_S (real-space lattice sum) plus Γ_L^{F} (mesh-free lattice Fourier sum, $k_c = 2\xi \cdot 6.5$) against Γ_{ref}. The line is the pair-level error $\max_r |\theta_S + \theta_L - e^{-\kappa r}/r|/(e^{-\kappa r}/r)$. Grey crosses: the Fourier sum against the real-space image sum Γ_L with $|\mathbf{k}|$ rounded to twelve decimals, the earlier coefficient routine (Table 2). (b) Real-space and (c) Fourier truncation at $\xi = 0.7$, with no other approximation present. The curves are the median over three seeds and the band is their range; dotted curves are cases A and B; the dashed curve is e^{-x^2}; the dotted horizontal line is the fit floor $100\,\varepsilon_{\mathrm{num}}$. (d) Mesh error against the mesh-free lattice sum over the same modes at the production $k_c = 5.74$, for uniform $N = 128$, seed 1. The curves are the RMS over eight grid shifts, the bands the range over shifts, and the thin segments have slope p. The dotted line is the fit floor; no point of this scan lies below it. (e) Fitted slopes in h on the four finest meshes above the fit floor. Filled markers use the production k_c; open markers use k_c at $s = 5.5$ (five configurations). (f) Largest over RMS error across the eight grid shifts, for every mesh and p of the eight-shift scans ($N \leq 128$, production k_c).

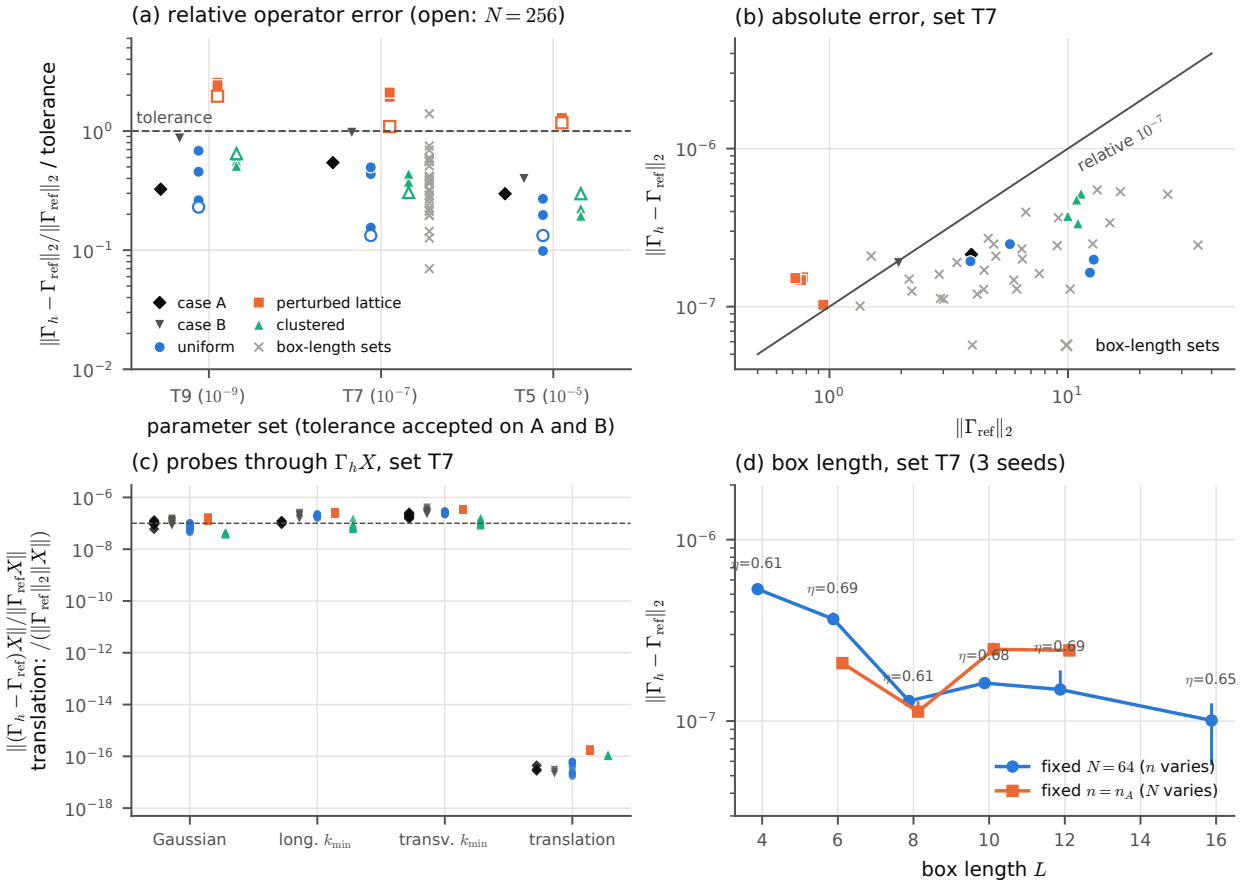


Figure 2: The production action $X \mapsto \Gamma_h X$ against the dense full-periodic reference. (a) Relative operator error divided by the tolerance for which the set was accepted on cases A and B (10^{-9}, 10^{-7}, 10^{-5}); the dashed line is the tolerance. Filled markers are $N \leq 128$, open markers $N = 256$, and grey crosses the box-length sets (T7 only). (b) Absolute error against $\|\Gamma_{\mathrm{ref}}\|_2$ at T7; the diagonal is relative 10^{-7}; crosses are the box-length sets. (c) Relative action errors of the probes through the production matvec at T7, main and legacy sets; the dashed line is 10^{-7}. For the translations the operator-normalized error $\|(\Gamma_h - \Gamma_{\mathrm{ref}})T_\alpha\|/(\|\Gamma_{\mathrm{ref}}\|_2\|T_\alpha\|)$ is shown; since $\Gamma_h T_\alpha = 0$ exactly, it equals the reference residual. (d) Absolute error against box length at T7, at fixed $N = 64$ (density varies) and at fixed density n_A (N varies). The realised η is annotated.